

Simulations with OpenFOAM

MPI initialization with
coupling and mesh definition

CFD
(PIMPLE/PISO/SIMPLE)
algorithm

Create sampled data $\mathcal{H}(\mathbf{u}_k^f)$

Send forecast velocity field $\mathbf{u}_k^f$
and parameters θ_k^f

Receive updated velocity field $\mathbf{u}_k^a$
and parameters θ_k^a

Coupling deletion with MPI
finalization

EnKF

MPI initialization with
coupling and mesh definition

Receive forecast velocity field $\mathbf{u}_k^f$
and parameters θ_k^f

Read sampled $\mathcal{H}(\mathbf{u}_k^f)$
and observation $\mathbf{y}_k$ data

EnKF algorithm

Send back updated velocity field $\mathbf{u}_k^a$
and parameters θ_k^a

Coupling deletion with MPI
finalization

Create observation
data $\mathbf{y}_k$

**DA algorithm
at time k**

